Tutorial Sheet 2

Memory Management

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PUBLISHED
16.05.2024

Introduction

1.1 The MMU

a) What are the tasks of memory management in operating systems?

b) What tasks does the MMU perform? What is the purpose of dividing a virtual address into segment number (or page number) and offset?

c) Is it possible that a virtual address is called for which no physical address exists? If so, how should the MMU handle it?

d) Why is it beneficial to implement the MMU as a hardware component rather than performing the necessary calculations in software?

1.2 System Performance

Given a demand paging system that is currently occupied as follows:

- CPU load: 20%
- ratio of page fault for accessing memory: 97,7%
- load of the other I/O devices: 5%

The aim is to optimize the CPU load in order to process all programs fast and efficiently. For each of the following suggestions, evaluate whether it would improve the CPU load (i.e. increase):

1. install a faster CPU
2. install a larger hard disk
3. increase the number of running/ready processes
4. reduce the number of running/ready processes
5. install more main memory
6. install a faster hard disk

2. Segmentation

For memory management using segmentation, the following segment table is given for a process:

segment	base address	length
`0x0`	`0x528`	320

segment	base address	length
`0x1`	`0x232`	58
`0x2`	`0x136`	120
`0x3`	`0x2ee`	60
`0x4`	`0x33e`	150
`0x5`	`0x3de`	110

Both the base address and the length are given in bytes.

a) How many bytes are available to the process in physical memory?

b) The address is coded on 16 bits. The maximum segment number is 16, how would you split the virtual address between the *segment number* and the *address* within the segment? What is the maximum segment length?

c) The process requests the address `0x31` within segment number 2, what virtual address will be sent to the MMU?

d) At what time is the virtual address from question **c)** determined?

e) If two separate processes each have a memory segment labeled with the same segment number within their address space, how does the operating system ensure that these segments do not overlap in physical memory?

f) The following virtual addresses are now requested by the process. What does the MMU return in each case?

- `0x301e`
- `0x1012`
- `0x207a`
- `0x4095`

3. Paging

Suppose we have a 32 bits system, managing virtual memory with paging. The page size is 4KiB and the TLB is empty.

The system uses a two level page table, the virtual addresses are divided as follows:

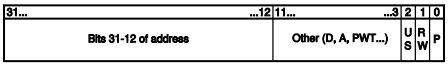
- 10 bits for the index of the first level page table
- 10 bits for the index of the second level page table
- 12 bits for the offset within the page

For instance, the address `0xc01b36e4` is divided like this: `0b1100000000|0110110011|011011100100`

- the index for the first level of the page table is 768
- the index for the second level is 435
- the offset is 1764

The Page Table Entry is composed of 20 bits for the page frame number, and the last 12 bits are used for metadata.

PTE



The last three bits represent the following flags:

- **U/S**: set to 1 if the page can be accessed either in user or supervisor mode
- **R/W**: set to 1 if the page is r/w, 0 if read-only
- **P**: set to 1 if the page is present in memory

Memory pages

1023	0x761c62fc
1022	0xec0fe000
1021	0x044e2fe1
…	…
610	0x3ed733d1
609	0x54826000
608	0xb0b41068
607	0x1432c000
…	…
382	0xc13e4000
381	0x5fc46348
…	…
7	0x94ed9556
6	0x771af000
5	0x554c9d87
4	0x6cb4f481
3	0x0c9a4000
2	0xd9ffb83a
1	0xf0b77ac6
0	0x321e7b42

`addr: 0xef2e5000`

1023	0xf0be9fcd
…	…
261	0xf75c68ae
262	0x5433c09f
263	0x53b40b50
264	0xb3556864
265	0xd3e37a21
266	0x5ce775f3
267	0xaee71050
268	0x84ab8b74
269	0x83cb8c1a
270	0x79ae8204
271	0x6b2ba370
…	…
5	0x3c3704fd
4	0x73839cc3
3	0xce6399e0
2	0xd1fd6562
1	0xcd1fe587
0	0xdd02d6e7

`addr: 0x771af000`

1023	0xe11d72a3
1022	0x486143b2
1021	0xcaa73b51
1020	0x74b23643
1019	0x8c88cbd3
1018	0x064a0623
1017	0x5a7e0b13
1016	0xb3ff3b53
1015	0xd2da3be1
1014	0x22abdd03
…	…
674	0xa7f0a152
673	0x5e55ba43
672	0x747a7a3f
671	0x61ae7da2
670	0xf29c9732
669	0x05cbf3a1
…	…
1	0x6cd67cb1
0	0x4215b193

`addr: 0x1432c000`

1023	0xbdd85d16
…	…
837	0x1384f7af
836	0xcdb54af2
835	0x07959e6c
834	0x45c0c51a
833	0x13db28dd
832	0x0d72ddd7
831	0x6eba6034
…	…
9	0x36556fd1
8	0x1fb73c5c
7	0xbcb8acb7
6	0xb975c46e
5	0x4830d301
4	0x59ff1322
3	0x5c6f1356
2	0x22da0265
1	0x65103266
0	0x3e797c52

`addr: 0x54826000`

1023	0x0ed71b28
…	…
979	0xe2931f71
978	0xbf84f8f6
977	0x212c5399
976	0x1fa440ac
975	0xc8b8dda1
974	0x73b5897a
973	0x64c2d6f9
972	0xe6532b3c
971	0xb0a368a1
970	0x910d0807
969	0xdf3f3877
968	0x45dc0a39
967	0x03b373e2
966	0xb869f0bd
…	…
2	0x322991db
1	0xc7051491
0	0x58f86463

`addr: 0xc13e4000`

Suppose that a task is running, the page table pointer of that task contains `0xef2e5000`.

Questions

1. The third page displayed above starts at the address `0x1432c000`. What is the associated page number?
2. The value `0x6cb4f481` is stored inside the first page, at the index 4. What is the virtual address at which the value is stored?
3. What is the size of each page table level? Why is it convenient?
4. What is the total size of memory used by the page table for a single address translation?

Task

The CPU is performing the following actions:

1. In user mode, read at the address `0x97ea0329`
2. In user mode, write to the address `0x5fbc6582`
3. In kernel mode, read at the address `0x5fbcd6f0`
4. In user mode, read at the address `0x5fbd2bad`
5. In user mode, read at the address `0x97ffbc21`
6. In user mode, read at the address `0x97ea0fe2`

For each action, document the steps involved in MMU translation. Explain how the MMU accesses the Page Table Entry associated with the virtual address, translates the virtual address to a physical address, and handles any interrupts or exceptions encountered during the translation process.

TLB

Translation Lookaside Buffer

Page	Page Frame	FLAGS

Finally, update the TLB accordingly: each entry one after the other. To make things easier, don't flush the TLB and suppose that all entries are valid (!= present).

> ⓘ **Note**
>
> On a more realistic system, the TLB contains more flags for caching etc.
>
> The TLB is either flushed between each context switch or contains a **PCID** (Process Context Identifier) to associate each entry with a specific process.

Bonus: Segmentation vs. Paging

Given a computer with **16 GiB** main memory and **64-bit** address space. You are considering whether to use segmentation or paging. You have the choice between 3 options.

Note: 1 GiB = 2^{30} bytes, 1 MiB = 2^{20} bytes

Option 1: Paging with pages of 4 bytes in size

1.1 How many entries would there be in the page table?

1.2 Why such a small page size could cause a lot of TLB misses.

1.3 What is the size of the page table for a single translation, with 1 level of page table?

1.4 What about 2 levels of page table?

1.5 What about 7 levels of page table?

1.6 Why is this number of levels problematic?

Option 2: Paging with pages of 256 MiB in size

2.1 How many entries would there be in the page table?

2.2 Why is having pages of this size problematic?

Option 3: Segmentation with best fit as a strategy for searching for free memory

3.1 Is there any internal / external fragmentation?

Conclusion:

Based on the questions discussed above, what can we conclude about the optimal solution for memory management?